



Graph-Augmented Retrieval-Augmented Generation for Vietnamese Labor Law Question Answering

Thesis Report

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Abstract

TODO: Write a concise 200–300 word abstract after chapter drafting. Mention the scoped Vietnamese labor-law corpus, hierarchy-aware chunking, Qdrant, Neo4j, graph-augmented retrieval, extractive citation-grounded answering, deterministic rule-based validation, and the constructed 69-query benchmark.

Acknowledgements

TODO: Add acknowledgements before submission.

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Chapter 1

Introduction

1.1 Background

TODO: Introduce Vietnamese labor law as a structured, citation-heavy information access domain. Explain why users need answers that preserve article, clause, point, document, and normative-rank context rather than only keyword matches.

1.2 Problem Statement

TODO: State the concrete problem: retrieval-only legal QA can miss cross-referenced provisions, implementing guidance, appendices, and procedure rules, which can lead to incomplete or weakly grounded answers.

1.3 Research Objectives

TODO: List measurable objectives: build a Vietnamese labor-law Graph-Augmented RAG pipeline, construct a hierarchy-aware corpus and graph, retrieve citation-grounded evidence, generate extractive answers with legal citations, and evaluate on the constructed 69-query benchmark.

1.4 Scope of the Project

TODO: Define the project scope around the six indexed Vietnamese labor-law and labor-procedure documents. Clarify that the system is for information assistance within the indexed corpus and is not a substitute for professional legal advice.

1.5 Contributions

TODO: Summarize contributions: cleaned legal corpus, hierarchy-aware legal chunking, metadata enrichment, cross-reference extraction, Qdrant vector index, Neo4j legal knowledge graph, graph-augmented retrieval, citation-grounded extractive answer generation, and deterministic validation artifacts.

1.6 Thesis Structure

TODO: Briefly describe Chapters 2–7 after drafting: background and related work, methodology, implementation, evaluation, discussion, and conclusion with future work.

Chapter 2

Background and Related Work

2.1 Large Language Models

TODO: Explain transformer-based language models, their use in question answering, and their limitations for legal domains, including hallucination, context limits, and the need for source grounding. Add verified citations.

2.2 Retrieval-Augmented Generation

TODO: Move and expand the old RAG background here. Explain indexing, embeddings, retrieval, context assembly, and grounded generation. Cite foundational RAG work such as [1] and add any newer verified RAG references.

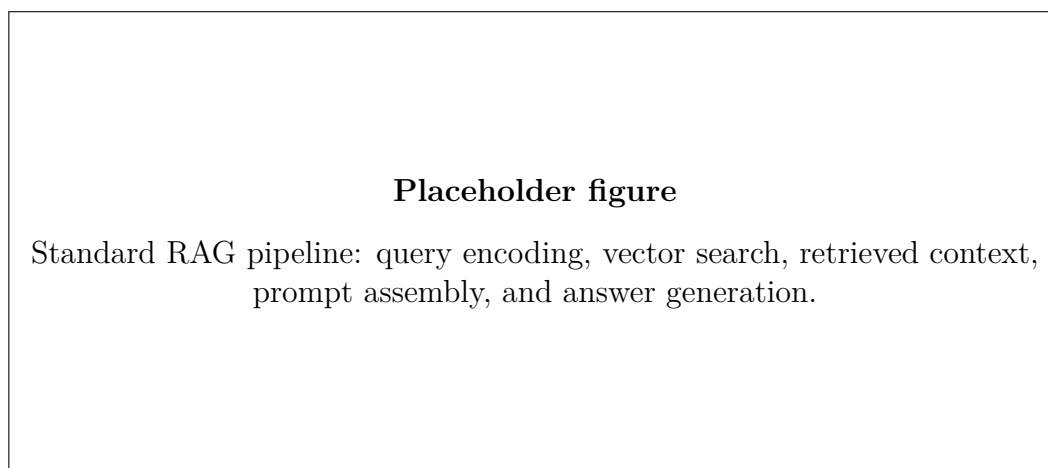


Figure 2.1: Standard retrieval-augmented generation pipeline.

2.3 Legal Question Answering

TODO: Review legal QA requirements: jurisdiction specificity, temporal validity, citation traceability, cautious wording, and the risk of unsupported legal conclusions. Add legal QA references using [4].

2.4 Legal Document Structure and Citation

TODO: Explain how Vietnamese legal documents are organized into documents, chapters, sections, articles, clauses, points, and appendices. Discuss why legal citations must preserve this hierarchy. Cite verified Vietnamese legal document sources using [2].

2.5 Knowledge Graphs for Legal Retrieval

TODO: Introduce knowledge graphs for representing legal hierarchy, references, topics, actors, issue types, and normative relationships. Explain how graph structure complements embedding similarity. Add verified KG and legal KG references.

2.6 Graph-Augmented Retrieval

TODO: Move the old graph-augmented retrieval background here. Compare graph expansion with vector-only and hybrid retrieval, and cite GraphRAG or graph-augmented retrieval work using [3].

2.7 Evaluation Methods for RAG Systems

TODO: Review retrieval metrics, answer validation, and citation grounding. Distinguish human expert review, LLM-as-Judge, and deterministic rule-based validation; later chapters use deterministic rule-based validation only.

Chapter 3

System Design and Methodology

3.1 Overall System Architecture

TODO: Move the old system architecture content here and revise it around the implemented Graph-Augmented RAG pipeline: corpus preparation, chunking, enrichment, cross-reference extraction, Qdrant indexing, Neo4j graph construction, graph-augmented retrieval, extractive answer generation, and rule-based citation validation.

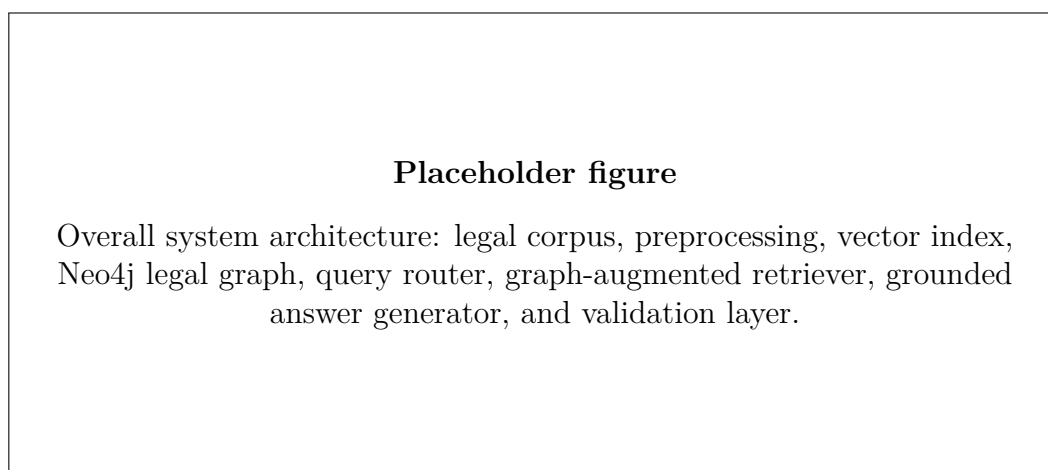


Figure 3.1: Overall system architecture.

3.2 Legal Corpus and Data Sources

TODO: Explain the intentionally scoped six-document corpus and state that it covers selected Vietnamese labor-law documents, implementing guidance, retirement appendices, and labor-related civil procedure provisions.

Table 3.1: Corpus document table.

Document ID	Type	Normative Rank	Chunks
45-2019-qh14	bo_luat	1	697
92-2015-qh13-labor-only	bo_luat	1	159
ngghi-dinh-135-2020-nd-cp	ngghi_dinh	2	41
ngghi-dinh-145-2020-nd-cp	ngghi_dinh	2	520
thong-tu-09-2020-tt-bltdtbxh	thong_tu	3	73
thong-tu-10-2020-tt-bltdtbxh	thong_tu	3	66

3.2.1 Labor Code 2019

TODO: Describe the role of the 2019 Labor Code in the corpus. Add official source details and final legal citation metadata before submission.

3.2.2 Decrees and Circulars

TODO: Describe how Decree 135/2020/ND-CP, Decree 145/2020/ND-CP, Circular 09/2020/TT-BLDTBXH, and Circular 10/2020/TT-BLDTBXH provide lower-rank implementation details and guidance.

3.2.3 Labor-related Civil Procedure Provisions

TODO: Explain why labor-related provisions from the Civil Procedure Code are included for labor dispute and court procedure questions. Clarify the filtering method and document scope.

3.3 Legal Document Preprocessing

TODO: Move old preprocessing content here and align it with the implemented artifact pipeline from cleaned text to validated legal chunks.

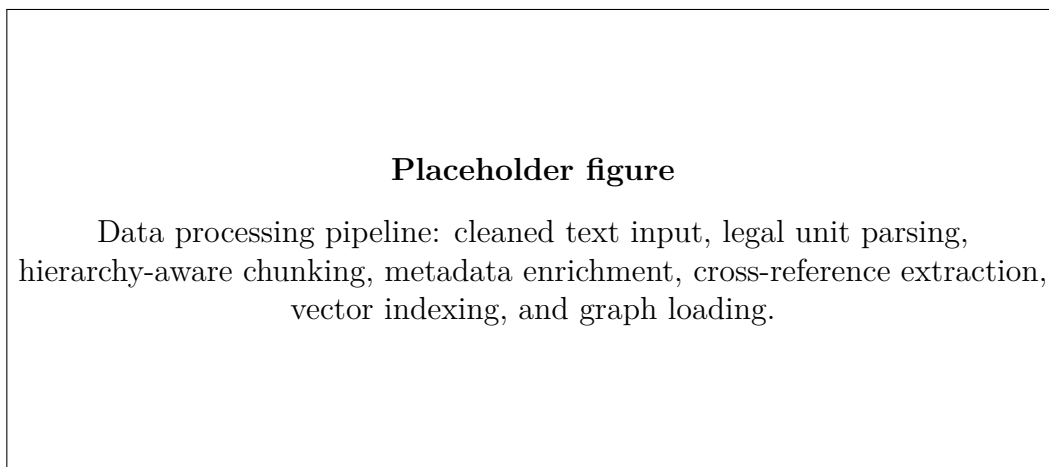


Figure 3.2: Data processing pipeline.

3.3.1 Text Cleaning

TODO: Explain cleaning of legal text inputs, encoding normalization, removal of extraction artifacts, and validation reports for curated text.

3.3.2 Hierarchical Chunking for Legal Texts

TODO: Explain the chunking algorithm for articles, clauses, points, appendices, and tables. Describe how citation text and parent hierarchy are preserved for retrieval and answer grounding.

Table 3.2: Chunking validation table.

Validation Item	Value
Legal chunks	1,556
Duplicate chunk IDs	0
Missing citation text	0
Missing article number	0
Very short chunks	0
Very long chunks	0

3.3.3 Appendix and Table Handling

TODO: Describe how appendix forms, appendix tables, and retirement-age lookup tables are represented as evidence chunks and graph nodes. Add examples in Appendix G.

3.3.4 Metadata Enrichment

TODO: Explain topic, actor, issue type, document type, normative rank, citation text, source file, and hierarchy metadata. Use counts from the final evaluation report when writing the full chapter.

3.4 Legal Cross-Reference Extraction

TODO: Describe the implemented extraction of legal references from chunk text into candidate graph edges. Include the resolved edge count of 948 and unresolved edge count of 111 without overclaiming perfect parsing.

3.4.1 Citation Pattern Detection

TODO: Explain citation patterns for article, clause, point, document-level references, and guidance relations. Add examples from the reference edge artifacts.

3.4.2 Document Alias Resolution

TODO: Explain how aliases such as Labor Code, decrees, circulars, and relative references are normalized to document IDs.

3.4.3 Reference Edge Generation

TODO: Describe generation and validation of REFERENCES, DETAILS, and GUIDED_BY edges before Neo4j loading.

3.5 Vector-based Retrieval

TODO: Describe the vector and hybrid retrieval baseline used before graph expansion. Keep the distinction between vector-only, hybrid, and graph-augmented modes clear.

3.5.1 Embedding Model

TODO: Document the dense embedding model, vector dimension 384, and why multilingual embeddings are suitable for Vietnamese legal text. Add a verified Sentence Transformers citation using [6].

3.5.2 Vector Index Payload Design

TODO: Explain the Qdrant payload fields, including chunk ID, document ID, document type, normative rank, citation text, dense text, sparse text, and source metadata. Add a Qdrant citation using [5].

3.5.3 Hybrid Retrieval

TODO: Describe how dense and sparse retrieval signals are combined for the hybrid baseline, including top-k and prefetch settings from the evaluation scripts.

3.6 Legal Knowledge Graph Design

TODO: Move the old Neo4j graph construction content here and split it into schema, structural edges, cross-reference edges, and normative hierarchy edges.

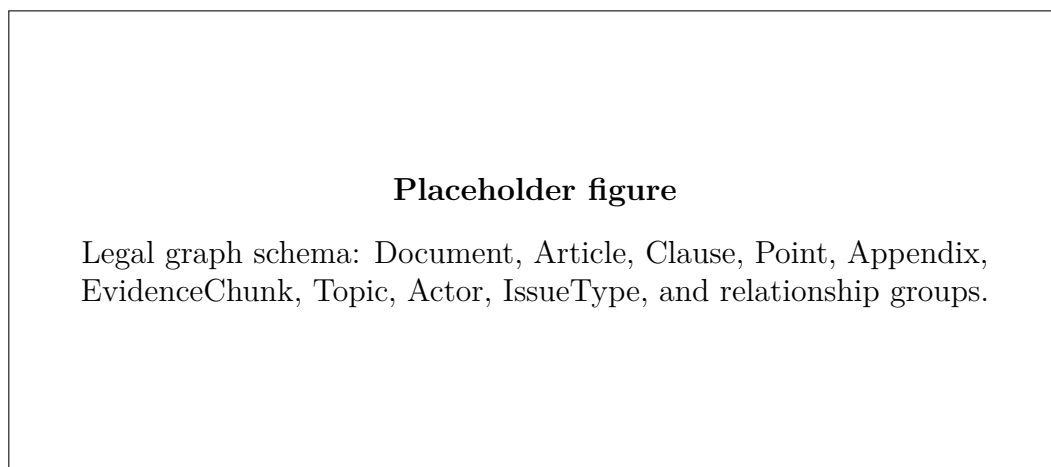


Figure 3.3: Legal graph schema.

Table 3.3: Graph node and edge count table.

Graph Item	Count
Documents	6
Articles	411
Clauses	1,235
Points	774
Appendices	35
Evidence chunks	1,556
Resolved reference edges loaded	948
Taxonomy edges	6,099
Normative hierarchy edges	20

3.6.1 Graph Schema

TODO: Explain the Neo4j node labels and key properties. Include constraints and indexes in Chapter 4.

3.6.2 Structural Edges

TODO: Describe document-to-article, article-to-clause, clause-to-point, appendix, and evidence-chunk relationships.

3.6.3 Cross-reference Edges

TODO: Describe loaded reference relationships and why unresolved references are excluded from graph traversal.

3.6.4 Normative Hierarchy Edges

TODO: Explain normative-rank relationships among codes, decrees, and circulars and how they support context ordering.

3.7 Query Routing and Retrieval Strategy

TODO: Explain how query intent and legal hints influence retrieval mode, metadata boosting, and graph expansion.

3.7.1 Query Intent Classification

TODO: Define intent categories such as direct QA, definition QA, document guidance, procedure QA, comparison QA, table lookup, exception-based QA, scenario QA, and multi-hop QA.

3.7.2 Intent-specific Retrieval Boosting

TODO: Explain how article mentions, document mentions, topic hints, actor hints, and issue-type hints affect retrieval scores or filters.

3.7.3 Graph Expansion Policy

TODO: Describe the graph expansion depth, maximum expanded chunks, relationship types, and deduplication policy used after seed retrieval.

3.8 Graph-Augmented Retrieval Pipeline

TODO: Describe the complete graph-augmented retrieval flow from user query to seed retrieval, graph traversal, context merging, reranking, and final context selection.

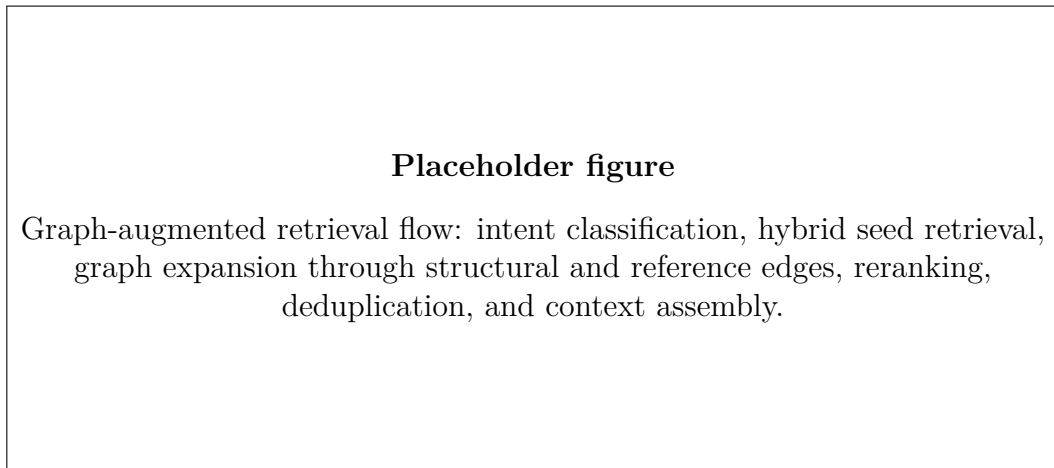


Figure 3.4: Graph-augmented retrieval flow.

3.9 Grounded Answer Generation with Legal Citations

TODO: Describe the extractive answer generator and validation layer. Avoid claiming broad legal reasoning beyond the retrieved and cited legal context.

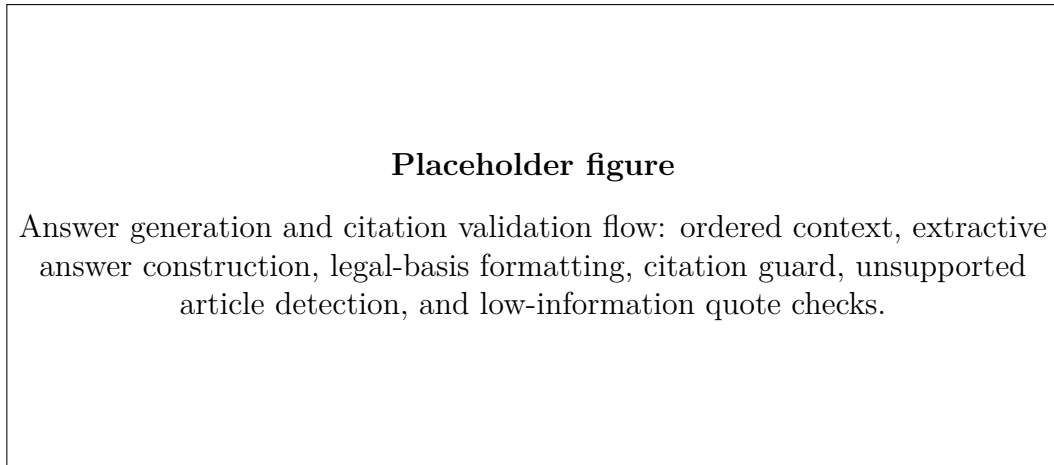


Figure 3.5: Answer generation and citation validation flow.

3.9.1 Context Ordering by Normative Rank

TODO: Explain how higher-rank legal documents, implementing decrees, circular guidance, and appendices are ordered for answer construction.

3.9.2 Citation-aware Answer Construction

TODO: Explain how answer text, evidence quotes, and legal-basis lists are built from retrieved chunks without inventing article numbers.

3.9.3 Rule-based Citation Validation

TODO: Explain deterministic validation of cited article numbers, retrieved citation coverage, forbidden citations, and low-information quotes.

Chapter 4

Implementation

4.1 Technology Stack

TODO: Move old implementation content here and update it for the actual project stack. Demote frontend, deployment, and CI details unless they are necessary for reproducing the evaluated RAG pipeline.

Table 4.1: Technology stack.

Layer	Technology	Purpose
Backend	Python, FastAPI	API and orchestration
Embeddings	Sentence Transformers	Dense multilingual vectors
Vector database	Qdrant	Dense and sparse retrieval
Graph database	Neo4j	Legal hierarchy and references
Evaluation	Python scripts and generated artifacts	Benchmark reporting

4.2 Data Processing Pipeline

TODO: Document the implemented scripts and modules that transform cleaned legal text into enriched chunks, reference edges, vector records, and graph artifacts.

4.2.1 Clean Text Input

TODO: Describe cleaned corpus inputs and validation artifacts from `artifacts/validation`.

4.2.2 Chunk Generation

TODO: Describe chunk generation modules and scripts, including hierarchy detection and output files `legal_chunks.jsonl` and `legal_chunks_summary.md`.

4.2.3 Metadata Enrichment

TODO: Describe enrichment rules for topic, actor, issue type, document type, normative rank, and citation metadata.

4.2.4 Cross-reference Extraction

TODO: Describe reference extraction outputs, including resolved and unresolved reference reports.

4.3 Vector Index Implementation

TODO: Move old vector database implementation content here and align it with the Qdrant collection used in the final evaluation.

4.3.1 Qdrant Collection

TODO: Document collection `vietnamese_labor_law_chunks`, vector dimension 384, cloud storage mode, and validation that 1,556 chunks were indexed.

4.3.2 Dense and Sparse Retrieval

TODO: Explain dense vector search, sparse token search, and how hybrid retrieval is implemented. Add key script and module names during chapter writing.

4.3.3 Metadata Payload

TODO: List payload fields and explain how they support filtering, citation display, reranking, and validation.

4.4 Neo4j Graph Implementation

TODO: Move old Neo4j graph database details here and keep lower-level implementation details separate from the methodology chapter.

4.4.1 Node Types

TODO: Document implemented node types: Document, Article, Clause, Point, Appendix, EvidenceChunk, Topic, Actor, and IssueType.

4.4.2 Relationship Types

TODO: Document structural, taxonomy, cross-reference, and normative hierarchy relationships. Include `REFERENCES`, `DETAILS`, and `GUIDED_BY`.

4.4.3 Graph Constraints and Indexes

TODO: Document Neo4j constraints, indexes, upsert strategy, and validation queries used during graph build.

4.5 Graph-Augmented Retriever Implementation

TODO: Describe implemented retriever modules and how they use Qdrant records plus Neo4j expansion.

4.5.1 Seed Chunk Selection

TODO: Explain seed retrieval from hybrid search, including top-k, prefetch limit, meta-data hints, and query routing outputs.

4.5.2 Graph Expansion

TODO: Explain graph expansion from seed chunks through structural, cross-reference, taxonomy, and normative hierarchy edges.

4.5.3 Reranking and Deduplication

TODO: Explain score merging, deduplication by chunk ID, context ordering, and maximum context selection for answering.

4.6 Grounded Answer Generator

TODO: Move old prompt design content here but revise it to match the implemented grounded extractive answer generator instead of a general free-form prompt-only design.

4.6.1 Extractive Answer Generation

TODO: Describe the deterministic extractive provider used in the final evaluation and how answer text is selected from retrieved evidence.

4.6.2 Legal Citation Formatting

TODO: Document legal-basis formatting, citation surfaces, evidence quotes, and source display.

4.6.3 Normative Hierarchy Handling

TODO: Explain how the answer generator handles higher-rank provisions, implementing guidance, and appendices without implying universal legal interpretation.

4.7 Evaluation Scripts and Artifacts

TODO: Document scripts for retrieval-mode evaluation, end-to-end evaluation, and final thesis evaluation reports. Reference the generated artifacts in `artifacts/evaluation`.

4.8 Example Use Cases

TODO: Move old example-use-case content here. Use the case studies requested in Chapter 5 as the main examples: minor worker, retirement age table lookup, labor contract content with circular guidance, and dismissal dispute with court procedure.

4.9 Reproducibility Setup

TODO: Move old deployment and CI/CD material here only if it helps reproduce the thesis evaluation. Include environment variables, Qdrant and Neo4j setup, and commands from `artifacts/evaluation/reproducibility_notes.md`.

Chapter 5

Evaluation and Results

5.1 Evaluation Setup

TODO: Introduce the evaluation package and reference `artifacts/evaluation/final_thesis_evaluation_report.md`. Do not add metrics beyond the generated artifacts.

5.1.1 Corpus Statistics

TODO: Summarize corpus, chunk, vector index, and graph statistics from the final evaluation report.

Table 5.1: Corpus and index statistics.

Item	Value
Corpus documents	6
Legal chunks	1,556
Vector index count	1,556
Neo4j documents	6
Neo4j articles	411
Neo4j clauses	1,235
Neo4j points	774
Neo4j appendices	35
Neo4j evidence chunks	1,556
Resolved reference edges loaded	948
Taxonomy edges	6,099
Normative hierarchy edges	20

5.1.2 Benchmark Construction

TODO: Describe construction of the 69-query benchmark, including query categories, required citation labels, forbidden citation checks, graph-required labels, difficulty labels, and normative-hierarchy-required labels.

5.1.3 Evaluation Environment

TODO: Document the evaluation environment from reproducibility notes: PowerShell on Windows, repository virtual environment, sentence-transformer embedding provider, Qdrant collection, Neo4j graph, top-k settings, and output artifacts.

5.2 Curated Legal QA Benchmark

TODO: Describe the curated benchmark design without implying it is exhaustive or expert-certified.

5.2.1 Query Categories

TODO: List query categories and counts from the benchmark: calculation/table lookup, comparison, definition, direct, document guidance, exception-based, multi-hop, procedure, and scenario-based QA.

5.2.2 Required Citation Labels

TODO: Explain that each benchmark item contains required citation labels used to evaluate retrieval coverage and answer citation grounding.

5.2.3 Forbidden Citation Checks

TODO: Explain forbidden citation checks and report that the final retrieval comparison found forbidden citation violation rate 0.000 for graph-augmented, hybrid, and vector-only modes.

5.2.4 Difficulty Levels

TODO: Describe easy, medium, and hard difficulty labels. The final benchmark contains 15 easy, 32 medium, and 22 hard queries.

5.3 Retrieval Evaluation

TODO: Define the retrieval evaluation as a comparison among vector-only, hybrid, and graph-augmented retrieval on the same 69-query benchmark.

5.3.1 Retrieval Modes

TODO: Describe vector-only retrieval, hybrid retrieval, and graph-augmented retrieval. Explain that graph-augmented retrieval starts from seed results and expands through the legal graph.

5.3.2 Retrieval Metrics

TODO: Define Recall@5, Recall@10, Precision@5, Precision@10, MRR, required citation coverage, and forbidden citation violation.

5.4 Answer Generation Evaluation

TODO: Describe the end-to-end evaluation of graph-augmented retrieval plus extractive answer generation.

5.4.1 Deterministic Rule-based Validation

Answer quality is evaluated using deterministic rule-based validation, not human legal expert review or LLM-as-Judge evaluation.

TODO: Explain the implemented deterministic checks and avoid using RAGAS or LLM-as-Judge terminology as final evaluation methodology unless clearly marked as future work.

5.4.2 Direct Answer Check

TODO: Explain how the validator checks that an answer directly addresses the benchmark question.

5.4.3 Required Legal Rule Check

TODO: Explain how required legal rules and required citations are checked against generated answer fields and cited retrieved contexts.

5.4.4 Low-information Answer Detection

TODO: Explain how low-information quotes are detected. The final evaluation reports low-information quotes: 0.

5.5 Citation Evaluation

TODO: Describe citation-specific checks used by the deterministic validator.

5.5.1 Citation Grounding

TODO: Explain that answer citations must be supported by retrieved contexts and mapped to known legal citation surfaces.

5.5.2 Unsupported Article Number Detection

TODO: Explain unsupported article number detection. The final evaluation reports unsupported article numbers: None.

5.5.3 Unretrieved Citation Detection

TODO: Explain unretrieved citation detection. The final evaluation reports unretrieved citations: None.

5.6 Retrieval Results

TODO: Insert and discuss retrieval comparison results from `retrieval_modes_expanded_report.md`.

5.6.1 Overall Retrieval Comparison

Table 5.2: Retrieval modes comparison table.

Mode	R@5	R@10	P@5	P@10	MRR	Coverage	Forbidden
Vector-only	0.012	0.012	0.026	0.020	0.022	0.012	0.000
Hybrid	0.860	0.920	0.571	0.380	0.833	0.920	0.000
Graph-augmented	0.961	1.000	0.562	0.359	0.872	1.000	0.000

TODO: Explain that graph-augmented retrieval improved Recall@10 from 0.920 to 1.000 compared with hybrid retrieval on the constructed benchmark, while precision decreased slightly because graph expansion adds supporting context.

5.6.2 Per-category Results

Table 5.3: Category-level result table.

Category	Q	Vec R@10	Hyb R@10	Graph R@10	Hyb Cov.	Graph Cov.
Calculation/table lookup	5	0.067	1.000	1.000	1.000	1.000
Comparison QA	4	0.000	1.000	1.000	1.000	1.000
Definition QA	7	0.000	1.000	1.000	1.000	1.000
Direct QA	19	0.000	1.000	1.000	1.000	1.000
Document guidance QA	10	0.000	0.683	1.000	0.683	1.000
Exception-based QA	11	0.045	0.955	1.000	0.955	1.000
Multi-hop QA	2	0.000	0.583	1.000	0.583	1.000
Procedure QA	7	0.000	0.857	1.000	0.857	1.000
Scenario-based QA	4	0.000	1.000	1.000	1.000	1.000

5.6.3 Graph-required Query Results

TODO: Report graph-required query results: 25 graph-required queries achieved 1.000 retrieval pass, answer pass, citation pass, end-to-end pass, and average quality score 100.00 in the final graph-augmented end-to-end run.

5.7 End-to-End Results

TODO: Report only the final generated metrics. Do not generalize the pass rate beyond the benchmark scope.

5.7.1 Overall End-to-End Pass Rate

Table 5.4: End-to-end evaluation summary table.

Metric	Value
Benchmark queries	69
End-to-end pass rate	1.000
Retrieval pass rate	1.000
Answer pass rate	1.000
Citation validation pass rate	1.000
Quality validation pass rate	1.000
Low-information quotes	0
Unsupported article numbers	None
Unretrieved citations	None

5.7.2 Category-level Results

TODO: Add category-level end-to-end pass rates from `final_thesis_evaluation_report.md`. The final report states that all evaluated topics reached 1.000 retrieval, answer, citation, and end-to-end pass rates in the graph-augmented run.

5.7.3 Difficulty-level Results

TODO: Report difficulty-level results: easy 15 queries, medium 32 queries, hard 22 queries; each group reached 1.000 retrieval pass, answer pass, citation pass, and end-to-end pass in the final graph-augmented run.

5.7.4 Normative-hierarchy-required Results

TODO: Report normative-hierarchy-required results: 22 true and 47 false queries; both groups reached 1.000 end-to-end pass rate in the final graph-augmented run.

5.8 Case Studies

TODO: Add concise case studies using actual retrieved contexts and generated answers from the final evaluation artifacts.

5.8.1 Minor Worker Question

TODO: Show how the system retrieves provisions for workers under 15 years old and cites the relevant Labor Code and guidance documents.

5.8.2 Retirement Age Table Lookup

TODO: Show how appendix or table chunks are retrieved and cited for retirement-age lookup questions.

5.8.3 Labor Contract Content with Circular Guidance

TODO: Show how graph expansion connects Labor Code contract provisions to circular guidance.

5.8.4 Dismissal Dispute and Court Procedure

TODO: Show how labor discipline or dismissal content is connected with labor-dispute procedure provisions.

5.9 Error Analysis and Limitations

The 100% result is measured on the constructed 69-query benchmark and should not be interpreted as universal legal correctness.

TODO: Discuss remaining limitations: scoped corpus, constructed benchmark, deterministic validation rather than expert review, extractive answer style, dependence on chunking and reference parsing quality, and external legal updates outside the indexed corpus.

Chapter 6

Discussion

6.1 Key Findings

TODO: Discuss findings supported by the generated evaluation artifacts: graph augmentation improved required citation retrieval over hybrid retrieval on the constructed benchmark, and the complete graph-augmented extractive pipeline passed deterministic validation checks.

6.2 Impact of Hierarchical Chunking

TODO: Explain how article, clause, point, and appendix-aware chunking affected evidence localization, citation formatting, retrieval filters, and validation.

6.3 Benefits of Graph-Augmented Retrieval

TODO: Discuss observed benefits for document-guidance, multi-hop, procedure, exception-based, and table-lookup queries. Use the category-level table from Chapter 5 as evidence.

6.4 Role of Normative Hierarchy in Legal QA

TODO: Explain the role of codes, decrees, circulars, and appendices in context ordering and retrieval expansion. Avoid claiming that rank ordering alone resolves legal conflicts.

6.5 Citation Grounding and Hallucination Control

TODO: Discuss how extractive generation and rule-based citation validation reduce unsupported article numbers and unretrieved citations within the benchmark scope.

6.6 Practical Applications

TODO: Discuss responsible uses for legal information access, HR support, study, and research. State that real legal decisions should be verified by qualified professionals.

6.7 Limitations

The 100% result is measured on the constructed 69-query benchmark and should not be interpreted as universal legal correctness.

Table 6.1: Limitations table.

Limitation	Thesis Treatment
Scoped corpus	State that the system covers six indexed legal documents and does not cover all Vietnamese law.
Constructed benchmark	Interpret 1.000 scores as benchmark-specific evidence only.
Deterministic validation	Clarify that answer quality was not reviewed by human legal experts or LLM-as-Judge.
Extractive answers	Explain that extractive answers improve grounding but may be less fluent or adaptive.
Upstream parsing dependence	Discuss how chunking, metadata, and reference extraction errors can affect retrieval.
Legal updates	Note that newer or external legal sources require corpus refresh and re-evaluation.

TODO: Expand this limitations table into prose after the final thesis argument is drafted.

Chapter 7

Conclusion and Future Work

7.1 Conclusion

TODO: Conclude with evidence-backed claims only: the implemented system integrates hierarchy-aware chunking, Qdrant, Neo4j, graph-augmented retrieval, extractive answer generation, and deterministic citation validation for a scoped Vietnamese labor-law corpus.

7.2 Future Work

TODO: Introduce concrete next steps that address the current benchmark and validation limitations.

7.2.1 Expert-validated Benchmark

TODO: Create a benchmark reviewed by Vietnamese labor-law experts and compare it with the current constructed benchmark.

7.2.2 LLM-as-Judge Evaluation

TODO: Add LLM-as-Judge evaluation only as a future complementary method with strict grounding checks, not as the current final evaluation method.

7.2.3 Human Legal Review

TODO: Evaluate answer usefulness, legal nuance, and risk with qualified legal reviewers.

7.2.4 Larger Legal Corpus

TODO: Expand the corpus to additional labor-related laws, implementing documents, official guidance, and newer amendments.

7.2.5 Improved Generative Answering

TODO: Explore constrained abstractive generation while preserving citation grounding, refusal behavior, and article-number validation.

7.2.6 Real-world Deployment

TODO: Study production requirements such as legal update monitoring, user feedback, audit logs, privacy, latency, and responsible-use warnings.

References

- [1] Patrick Lewis et al. “Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks”. In: *Advances in Neural Information Processing Systems*. 2020.
- [2] TODO. *TODO: Verified official sources for Vietnamese labor-law legal documents*. Placeholder citation key. Replace with official legal-document citation metadata before final submission.
- [3] TODO. *TODO: Verified reference for GraphRAG or graph-augmented retrieval*. Placeholder citation key. Replace with verified bibliographic metadata before final submission.
- [4] TODO. *TODO: Verified reference for legal question answering*. Placeholder citation key. Replace with verified bibliographic metadata before final submission.
- [5] TODO. *TODO: Verified reference for Qdrant vector database*. Placeholder citation key. Replace with verified bibliographic metadata before final submission.
- [6] TODO. *TODO: Verified reference for Sentence Transformers*. Placeholder citation key. Replace with verified bibliographic metadata before final submission.

Appendix A

List of Legal Documents

TODO: List the six legal documents used in the final corpus, including official titles, document IDs, effective dates, source URLs, document type, normative rank, and chunk counts.

Table A.1: Appendix document list placeholder.

Document ID	Official Title	Source URL	Notes
45-2019-qh14	TODO	TODO	Labor Code 2019
92-2015-qh13-labor-only	TODO	TODO	Labor-related procedure provisions
ngghi-dinh-135-2020-nd-cp	TODO	TODO	Retirement age guidance
ngghi-dinh-145-2020-nd-cp	TODO	TODO	Labor Code implementation
thong-tu-09-2020-tt-bldtbxh	TODO	TODO	Circular guidance
thong-tu-10-2020-tt-bldtbxh	TODO	TODO	Circular guidance

Appendix B

Chunk Schema

TODO: Document the JSONL schema for legal chunks and enriched chunks, including required fields, optional fields, and validation rules.

Appendix C

Graph Schema

TODO: Document Neo4j labels, relationship types, properties, constraints, indexes, and example Cypher queries.

Appendix D

Benchmark Samples

TODO: Add representative benchmark samples with question text, category, difficulty, required citations, forbidden citations, graph-required label, and normative-hierarchy-required label. Do not include fabricated examples.

Appendix E

Evaluation Metrics

TODO: Define retrieval metrics, answer validation checks, citation validation checks, and pass/fail aggregation. Include formulas used by the evaluation scripts.

Appendix F

Reproducibility Commands

TODO: Copy final reproducibility commands from `artifacts/evaluation/reproducibility_notes.md`, including retrieval-mode evaluation, end-to-end evaluation, optional vector index rebuild, and optional Neo4j graph rebuild.

Appendix G

Example Retrieved Contexts and Answers

TODO: Insert selected retrieved contexts and generated answers for the Chapter 5 case studies after choosing final examples from the evaluation outputs.